



RESEARCH ARTICLE

Invasion of rocky shores by a mytilid mussel reveals an abundant-centre distribution coupled with moderate increases in densities at its absolute range limits

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Abstract

Semimytilus patagonicus is an invasive mussel on the coast of southern Africa and has extended its range in recent years. We asked whether its distribution and abundance are consistent with the abundant-centre hypothesis (ACH). Marginal populations were located by monitoring 33 rocky shore sites in South Africa and southern Namibia in 2021. This revealed no changes to its distributional limits since 2020. At nine of these sites, population demography was measured to allow a comparison of their densities and size structure. Four were central populations on the west coast of South Africa (including the site where the species was first detected in 2009). Four were marginal populations in South Africa: two towards the cold range edge in the north and two towards the warm range edge to the south. The ninth population was in southern Namibia, representing a recent invasion event first detected in 2014. Across the species' South African range, the distribution of its abundance was generally consistent with the ACH, with the greatest abundance at its range centre and a gradual decrease towards the range edges. However, the ultimate marginal population at both its cold and warm range edges showed moderate upticks in abundance compared to the penultimate marginal populations. Additionally, marginal populations in South Africa typically included a greater proportion of large individuals. Recruitment intensity was greater in warm range edge populations than cold range edge populations. The size structure of the population in Namibia resembled those of central populations in South Africa. Moderate increases in densities at the absolute range limits suggest that the species is currently undergoing spread into regions associated with moderately optimal environmental conditions (ultimate range edge sites) after encountering regions associated with suboptimal environmental conditions (penultimate range edge sites).

KEYWORDS

alien species, Bivalvia, monitoring, Namibia, range-edge dynamics, recruitment, South Africa

INTRODUCTION

The abundant-centre hypothesis (ACH; Hengeveld & Haeck, 1982; Brown, 1984; Brown et al., 1995) predicts

high abundance in central populations of a species and that the species' population dynamics will differ fundamentally between its range edges and range centre. Monotonic decreases in abundance towards the range

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limits could be a result of marginal populations encountering competitive exclusion and/or suboptimal environmental conditions that physiologically limit population growth by suppressing reproductive output and/or recruitment success by reducing post-recruitment survival (Brown et al., 1995; Gilman, 2006; Sexton et al., 2009). In contrast, the range centre is expected to correspond to optimal habitat, leading to higher fitness. However, only a minority (39%) of empirical studies based on site occupancy and demographic parameters reviewed in Sagarin and Gaines (2002a) supported the abundant-centre distribution, suggesting that this biogeographical principle may not be universal (also see Baldanzi et al., 2013; Abeli et al., 2014; Dallas et al., 2017; Freeman & Beehler, 2018).

One defining feature of an abundant-centre distribution is lower genetic diversity for marginal populations than central ones because of the Allee effect on smaller populations and/or restricted gene flow in geographically isolated populations (Grayson & Johnson, 2017; Klopstein et al., 2006; Kottler et al., 2021; Lee, 2002). Despite the lack of generalization for the ACH at the population level (Sagarin & Gaines, 2002a), the majority of empirical studies on the predictions for genetic diversity (64%) reviewed in Eckert et al. (2008) supported the ACH. In particular, restricted gene flow (e.g. in short-dispersing species) can affect differences in genetic diversity between marginal and central populations, with lower rates of gene flow towards the range edges and higher flow towards range centres (Eckert et al., 2008; Kottler et al., 2021).

Lower genetic diversity at range edges in short-dispersing species may not always be explained by dispersal capacity or other demographic factors, however, but may instead be explained by a combination of other biotic elements intrinsic to the species and/or abiotic factors such as seasonality in temperature (San Juan et al., 2021). Randomly dispersing and long-distance dispersing species, such as highly mobile species and marine species with planktonic larvae (e.g. pelagic fishes, barnacles, mussels and sea urchins), are expected to be associated with higher levels of gene flow from geographically central to marginal sites (Bohonak, 1999; Eckert et al., 2008). This, in turn, is expected to weaken any genetic structure and reduce the genetic diversity of marginal populations, while minimizing negative outcomes such as inbreeding and local extinction (e.g. Consuegra et al., 2005; Ntuli et al., 2020). Limited evidence for the ACH has been reported for densities of marine species on rocky shores (Rivadeneira et al., 2010; Sagarin & Gaines, 2002b; Tam & Scrosati, 2011). Along the north-west Atlantic coast, intertidal mussels from wave-exposed sites exhibited an abundant-centre pattern with peak densities shifted further south from their geographic range centres (i.e. a ramped distribution; Tam & Scrosati, 2011). For another intertidal mussel, its spatial distribution in density was instead ramped north (Sagarin & Gaines, 2002b). These ramped (and also bimodal) distributional patterns, which deviate from the normal abundant-centre distribution predicted by the ACH, have been attributed to sudden changes in environmental conditions that affect physiological performance (Sagarin & Gaines, 2002b; Tam & Scrosati, 2011).

Efforts to reconstruct invasion histories of two intertidal alien mussels – namely, *M. galloprovincialis* and *Semimytilus patagonicus* (d'Orbigny, 1842), the latter species being previously known as *Semimytilus algosus* (Gould, 1850) – were undertaken (Ma, Gusha, et al., 2021; Ma, Zardi, et al., 2020; Signorelli & Pastorino, 2021). These efforts revealed that the majority of occurrence records (90%) of *M. galloprovincialis* were made a decade after the first successful introduction in 1979 in South Africa (Grant et al., 1984; Grant & Cherry, 1985; Ma, Gusha, et al., 2021) and that at least 28 years elapsed between the first and second occurrence records of another intertidal mussel, *S. patagonicus*, which invaded central Namibia

and later spread into southern Angola (Barnard, 1964; Lamy, 1931; Ma, Zardi, et al., 2020). The dearth of information during the initial years of their invasions of southern Africa obscures fine-scale patterns of spread. In a separate invasion event, surprisingly little is known of the early years of the invasion of South Africa by *S. patagonicus* despite its more recent discovery in 2009 (de Greef et al., 2013; Ma, McQuaid, et al., 2020; Robinson et al., 2020). Already by 2010, a large-scale survey of its distribution revealed that it had spread across 500 km of the South African west coast with high densities at Elandsbaai (where the species was first detected) and Yzerfontein, about 240 km south (de Greef et al., 2013). This extensive spatial range only 1 year after its detection and its absence from earlier intertidal studies in the region suggest that it has been introduced even earlier than 2009, perhaps within 10 years (de Greef et al., 2013). As predicted by Alexander et al. (2015), *S. patagonicus* has been spreading further south-east along the south coast of South Africa (Ma, McQuaid, et al., 2020; Robinson et al., 2020). Between 2010 and 2020, more than 340 km of the range was added to its distribution at an average rate of 7.5–40 km per year (Ma, McQuaid, et al., 2020).

For one of the more salient mytilid invasions of southern Africa, a monotonic decrease in abundance towards its warm range limit was observed recently, probably because of increasingly suboptimal environmental conditions as the species approaches a biogeographic transitional boundary, which acts as a strong barrier to spread (Assis et al., 2015; Ma, Gusha, et al., 2021). However, the range edge dynamics of this mytilid invasion have remained stable, and little is known of such dynamics at an active invasion front of mussels. Therefore, a closer examination of the range edge dynamics of the ongoing invasion by *S. patagonicus* can provide rare insights into the early stages of mytilid mussel invasions. The alien *S. algosus* has been detected in the coastal waters of Namibia for more than 90 years and in South Africa for more than a decade, and past monitoring has revealed that a distributional gap between these two invasion events had decreased substantially over time (de Greef et al., 2013; Lamy, 1931; Ma, McQuaid, et al., 2020; Ma, Zardi, et al., 2020). The contemporary range limits of the species in both countries have been determined at a relatively fine spatial resolution because of high sampling effort via targeted monitoring (Ma, McQuaid, et al., 2020) and biodiversity monitoring of rocky shores (Kreiner, Bolton, et al., 2019; Kreiner, Branch, et al., 2019; Kreiner, Reddy, et al., 2019). Here, we investigated whether the distribution and abundance of *S. patagonicus* are consistent with the abundant-centre distribution predicted by the ACH. To do so, we determined where its most recent absolute range limits are located in South Africa given that it has been expanding its range in recent times. Next, we made a demographic study (densities and size structure) of marginal and central populations to infer range edge dynamics and assess how well this agrees with the predictions of the ACH. Opportunistically, shell degradation, a proxy for endolithic infestation, was also estimated.

METHODS

Large-scale mussel monitoring

Between March and October 2021, we surveyed for the presence of *Semimytilus patagonicus* (the focal species) and other mussels during periods of spring low tides at 28 rocky-shore sites in South Africa (from Port Nolloth to Mosselbaai) and 5 sites in southern Namibia (from Hostel to Mittag; Figure 1). Since these regions in South Africa were surveyed

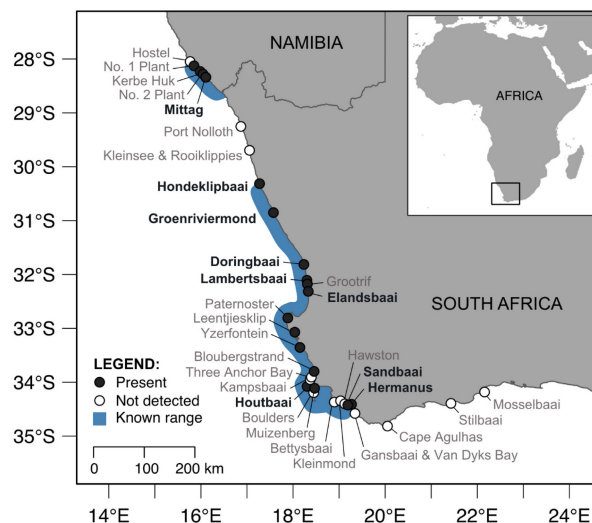


FIGURE 1 Occurrences of *Semimytilus patagonicus* from 28 rocky-shore sites in South Africa and 5 sites in southern Namibia in 2021; the known range is based on occurrence data from Ma, McQuaid, et al. (2020), Ma, Zardi, et al. (2020); names of sites at which sampling with quadrats were carried out are bold.

for *S. patagonicus* in early 2020 (Ma, McQuaid, et al., 2020), sampling was extended further north to Port Nolloth on the west coast and sampling effort was increased on the south coast in the vicinity of Hermanus. Opportunistically, the presence of other non-target mussel species in the intertidal zone – namely the Indigenous *Aulacomya atra* (Molina, 1782), *Choromytilus meridionalis* (F. Krauss, 1848) and *Perna perna*, plus the introduced *Mytilus galloprovincialis* – was also recorded (see Table S1).

Sampling with quadrats

Densities

Between August and October 2021, densities of *S. patagonicus* in mixed-species mussel beds were determined in situ for four marginal populations (Hondeklipbaai, Groenriviermond, Sandbaai and Hermanus), four central populations (Doringbaai, Lambertsbaai, Elandsbaai and Houtbaai) and one population representing a recent introduction in southern Namibia (Mittag; Figure 1) using 25 by 25 cm quadrats. Quadrats were placed haphazardly in the low balanoid zone above the cochlear zone and below the mid balanoid zone, where *S. patagonicus* is typically found (i.e. 0–40 cm above mean low water springs [MLWS]; de Greef et al., 2013). Twenty replicate quadrats were collected at each South African site and eight for the single Namibian site. The lower number of quadrats at the Namibian site was due to the logistical constraints of the remoteness of the site and the need to count substantially higher numbers of *S. patagonicus* over the course of a single low tide. The densities of new recruits of the year (i.e. ≤ 10 mm; based on growth rates reported by Zeeman et al., 2018) and post-recruitment individuals (> 10 mm) were estimated from size–frequency distribution data (see below). For the two sites with especially high levels of recruitment (Mittag and Elandsbaai), the densities of new recruits of the year were probably underestimated in the field because mussel length of ca. 1 mm was the lower limit of detection and the quadrat size may have been too large to sample mussel spat density accurately.

Per cent cover in southern Namibia

Per cent cover data were available only from southern Namibia as part of long-term monitoring since 2007 (Pulfrich, 2018, 2021). The mean per cent cover of *S. patagonicus* (comprising all size-classes, including mussel spat) in mixed-species mussel beds was determined for five populations (rocky shore sites: Hostel, No. 1 Plant, Kerbe Huk, No. 2 Plant, and Mittag; Figure 1) in 2015 to 2018 and in 2021 using 100 by 50 cm quadrats (Table 1). At each rocky shore site, sampling was done along five transects positioned perpendicular to the shore. Transects were spaced approximately 5–10 m apart at each site and six quadrats were placed at roughly equidistant intervals along each transect from the mean low water springs [MLWS] shore height to mean high water springs [MHWS], which ensured equivalent comparisons among sites. To assess triannual trends using the available data, rates of change (i.e. the percentage change in per cent cover between 2015 and 2018 and between 2018 and 2021) for each site were calculated using per cent cover values determined in 2015, 2018 and 2021 (Pulfrich, 2018, 2021).

Collection of animals

At three sites with high abundance (Mittag, Lambertsbaai and Elandsbaai), a representative sample of the *S. patagonicus* population was collected from a subset of quadrats made in 2021. At six other sites (Hondekliipbaai, Groenriviermond, Doringbaai, Houtbaai, Sandbaai and Hermanus), all specimens from the 20 quadrats (see above) and additional specimens located during a 45-min search for the species in the low balanoid zone were collected. Shell lengths of mussels were determined to the nearest 1 mm with a handheld Vernier calliper. The degree of shell infestation by endolithic cyanobacteria was estimated using a semi-quantitative approach by categorizing shells according to the percentage of the shell surface degraded following Ma, Gusha, et al. (2021). The categories were as follows: 0, >0 to 25, >25 to 50, >50 to 75, >75 to <100 and 100. For statistical analyses, estimates of shell degradation were converted into an index from 1 to 6, where an index of 1 is the lowest degree of degradation and 6 is the highest. For mussels collected from Mittag in Namibia, the extent of shell degradation was only estimated for a subset of mussels.

Data analyses

Shell length (continuous variable) and shell degradation index (categorical variable) were not tested in the same statistical model because these two

TABLE 1 Temporal trends in mean per cent cover (± 1 standard deviation) of *Semimytilus patagonicus* surveyed annually across five rocky shore sites in southern Namibia

Site	2007 to 2013	2014	2015	2016	2017	2018	2019	2020	2021
Hostel	0	0	0	0	0.01 $\pm$ 0.02	0.07 $\pm$ 0.10	n.d.	n.d.	0
No. 1 Plant	0	0	0.75 $\pm$ 1.05	0.01 $\pm$ 0.02	0	0.27 $\pm$ 0.23	n.d.	n.d.	10.13 $\pm$ 1.35
Kerbe Huk	0	0	0	10.64 $\pm$ 4.28	18.56 $\pm$ 3.41	1.67 $\pm$ 0.74	n.d.	n.d.	19.01 $\pm$ 6.41
No. 2 Plant	0	0	0.04 $\pm$ 0.82	7.43 $\pm$ 5.12	8.62 $\pm$ 8.27	5.91 $\pm$ 6.45	n.d.	n.d.	21.90 $\pm$ 5.52
Mittag	0	present	11.05 $\pm$ 4.80	5.79 $\pm$ 4.29	6.68 $\pm$ 4.25	9.68 $\pm$ 4.08	n.d.	n.d.	22.67 $\pm$ 4.54

Abbreviation: n.d., no data.

variables were correlated (examined using Spearman's rank-correlation coefficient). Among-site variability in the number of *S. patagonicus* individuals per quadrat, in shell length and in shell degradation index was determined using one-way analyses of variance (ANOVA) generalized linear model (GzLM) procedure with a Poisson error distribution and log-link function. For models that were statistically significant, a post hoc multiple-pairwise comparisons test was applied to the data to identify the different sites (*glht* function, package *multcomp*; Hothorn et al., 2008).

Abundant-centre patterns in South Africa and southern Namibia were evaluated by applying linear regressions on: (a) mussel density (response variable) and distance from the species' abundance centre at Elandsbaai (Figure 1) and the geographic range centre located about 160 km south of Elandsbaai near Saldanha Bay (explanatory variables), and (b) per cent cover (response variable) and distance from its abundance centre located at Mittag (explanatory variable; Figure 1), which is also the range centre (range: Hostel to Alexanderbaai; Ma, McQuaid, et al., 2020; Ma, Zardi, et al., 2020), for each surveyed year (following Yancovitch Shalom et al., 2020). Given the fractal nature of the coastline, distances were calculated by measuring at 1 km intervals (see Method in Ma, Redelinghuys, et al., 2021). These analyses (Spearman's rank-correlation coefficient, one-way ANOVAs using the GzLM procedure and linear regressions) were performed in the R programming environment (R Core Team, 2020).

RESULTS

Monitoring

Although previously detected at all five sites sampled in southern Namibia in 2021, *Semimytilus patagonicus* was found only at the four most southerly rocky shore sites (Figure 1). In South Africa, the species was detected from 14 sites ranging from Hondeklipbaai (i.e. cold-edge range limit) to Hermanus (i.e. warm-edge range limit; Figure 1). Within this geographic range, the species was not detected at six sites, concentrated across the south–south-eastern region of its distribution (i.e. south–south-east of Bloubergstrand; Figure 1).

Densities

Mean densities (± 1 standard deviation) of central populations of *S. patagonicus* in 2021 were $34\,258 \pm 12\,988 \text{ m}^{-2}$ at Mittag, $4.8 \pm 9.1 \text{ m}^{-2}$ at Doringbaai, $128 \pm 207 \text{ m}^{-2}$ at Lambertsbaai, $3\,794 \pm 6\,744 \text{ m}^{-2}$ Elandsbaai and $121 \pm 119 \text{ m}^{-2}$ Houtbaai. At marginal sites, mean densities were $24 \pm 16 \text{ m}^{-2}$ at Hondeklipbaai, $3.2 \pm 8.4 \text{ m}^{-2}$ at Groenriviermond, $4.0 \pm 4.9 \text{ m}^{-2}$ at Sandbaai and $262 \pm 735 \text{ m}^{-2}$ at Hermanus. Significant variability in *S. patagonicus* abundance was found among the nine rocky shore sites (one-way ANOVA using the GzLM procedure, $df = 8$, $p < 0.001$). Based on the post hoc multiple comparisons test, mussel abundance at the two penultimate sites (Groenriviermond and Sandbaai) was significantly different from all other sites. From the South African range, mussel densities were negatively correlated with distance from the abundance centre located at Elandsbaai (linear regression, $R^2 = 0.039$, slope = -2.362 , $p = 0.012$) and distance from the geographic range centre (linear regression, $R^2 = 0.049$, slope = -5.214 , $p = 0.005$). These two linear regressions are statistically significant even after applying a Bonferroni correction of the alpha level to 0.025.

Qualitatively, the estimated densities of new recruits of the year (≤ 10 mm), the estimated densities of post-recruitment individuals (> 10 mm) and the mean densities of individuals of all size-classes of *S. patagonicus* peaked at the centre site (i.e. Elandsbaai) and gradually decreased from this site to the penultimate marginal sites (Figure 2). At the species' ultimate warm range edge marginal site at Hermanus, densities of new recruits of the year increased relative to the nearby penultimate site at Sandbaai (Figure 2a), which was not the case at the cold range edge (i.e. Hondeklipbaai vs. Groenriviermond). Densities of post-recruitment individuals at ultimate marginal sites at its cold range edge at Hondeklipbaai and warm range edge at Hermanus increased relative to penultimate sites at Groenriviermond and Sandbaai respectively (Figure 2b). Densities of new recruits of the year and post-recruitment individuals from the recently introduced population in Namibia (Mittag) were substantially higher than at any of the South African populations (Figure 2).

Per cent cover in southern Namibia

During the earlier period between 2015 and 2018, rate of change in mean per cent cover varied substantially across sites, i.e. -12% at Mittag, 16029% at No. 2 Plant and -64% at No. 1 Plant. Because the species was not detected in 2015 at Kerbe Huk and Hostel, rates of change cannot be computed for these two sites for this time period. However, by 2018, the per cent cover of *S. patagonicus* decreased by an order of magnitude after 2 years (i.e. first detected in 2016) at Kerbe Huk and remained within the same order of magnitude after 1 year (first detected in 2017) at Hostel (Table 1). During the more recent period between 2018 and 2021, the rate of change in mean per cent cover increased with increasing distance from Mittag: 134% at Mittag, 271% at No. 2 Plant, 1040% at Kerbe Huk and 3604% at No. 1 Plant. Although the per cent cover of *S. patagonicus* was $< 1\%$ at Hostel in 2018, it was not observed in any quadrats in 2021, resulting in a rate of change in per cent cover of -100% over this time period (Table 1). To date, the mean *S. patagonicus* per cent cover (± 1 standard deviation) in southern Namibia in 2021 decreased with increasing distance from Mittag: $22.7 \pm 4.5\%$ at Mittag, $21.9 \pm 5.5\%$ at No. 2 Plant, 19.0 ± 6.4 at Kerbe Huk, $10.1 \pm 1.3\%$ at No. 1 Plant and 0% (i.e. not detected) at Hostel. Per cent covers were negatively correlated with distance from abundance centre located at Mittag, which is also the contemporary geographic range centre for each of the surveyed years: 2015 (linear regression, $R^2 = 0.314$, slope = -0.157 , $p = 0.004$), 2016 ($R^2 = 0.345$, slope = -0.182 , $p = 0.002$), 2017 ($R^2 = 0.243$, slope = -0.229 , $p = 0.012$), 2018 ($R^2 = 0.474$, slope = -0.195 , $p < 0.001$) and 2021 ($R^2 = 0.787$, slope = -0.493 , $p < 0.001$).

Population size structures

Despite geographic variation in densities across the South African range, the proportion of the *S. patagonicus* shell lengths shifts from mostly small-sized individuals in central populations to mostly larger-sized individuals in marginal populations at both the species' cold and warm range edges (Figure 3). The size–frequency distribution of the Namibian population, however, consisted mostly of smaller individuals (Figure 3). Overall, significant among-site variability in shell lengths was found (one-way ANOVA using the GzLM procedure, $df = 8$, $p < 0.001$) and post hoc tests revealed that shell lengths at the two ultimate sites (Hondeklipbaai and Hermanus) differed significantly from all other sites.

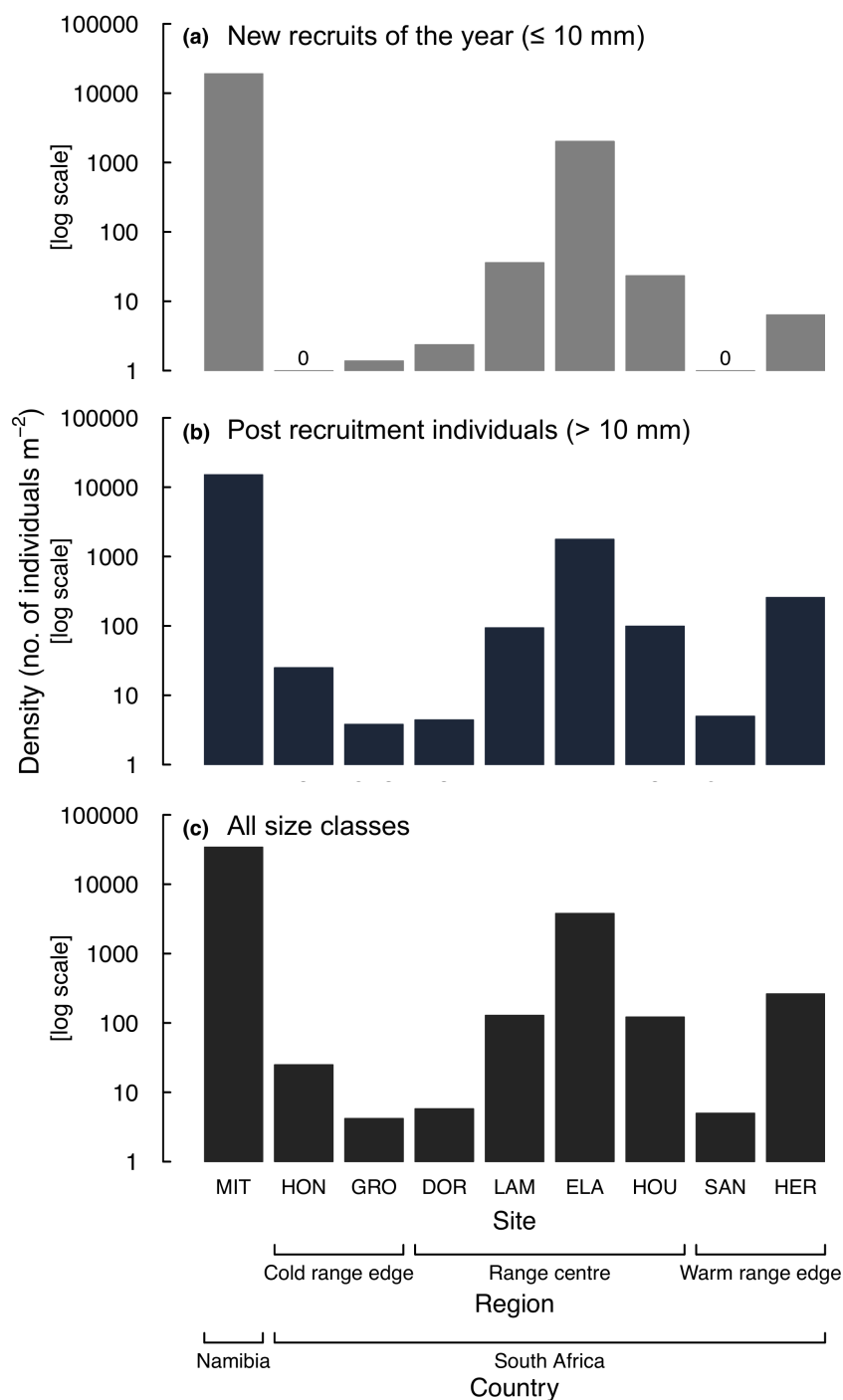


FIGURE 2 Among-site variation in densities of *Semimytilus patagonicus* populations from nine rocky shore sites (three cold range edge sites, four range centre sites and two warm range edge sites): (a) estimated densities of new recruits of the year (≤ 10 mm), (b) estimated densities of post-recruitment individuals (> 10 mm) and (c) mean densities of all size-classes; DOR, Doringbaai; ELA, Elandsbaai; GRO, Groenriviermond; HER, Hermanus; HON, Hondeklipbaai; HOU, Houtbaai (Hout Bay); LAM, Lambertsbai; MIT, Mittag; SAN, Sandbaai. Density values are the number of individuals m^{-2} ; note the scale is $\log_{10}$.

Shell degradation estimates

The extent of shell degradation in *S. patagonicus* increased with an increase in the size of the mussel (Spearman's rank-correlation coefficient,

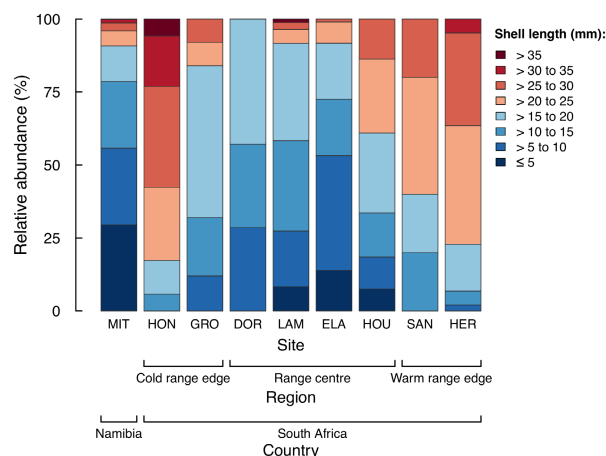


FIGURE 3 Size–frequency distribution of *Semimytilus patagonicus* populations from nine rocky shore sites (three cold range edge sites, four range centre sites and two warm range edge sites); MIT = Mittag ($n = 815$ individuals), HON = Hondeklipbaai ($n = 52$), GRO = Groenriviermond ($n = 25$), DOR = Doringbaai ($n = 7$), LAM = Lambertsbaai ($n = 84$), ELA = Elandsbaai ($n = 904$), HOU = Houtbaai (Hout Bay; $n = 146$), SAN = Sandbaai ($n = 5$) and HER = Hermanus ($n = 293$).

$\rho = 0.658$, $p < 0.001$). Across its South African range and for all size-classes, populations towards the warm-range edge tended to exhibit less shell degradation than sites towards the cold-range edge (Figure S1). Compared to the South African populations, the extent of shell degradation in the single Namibian site tended to be less, with most individuals being free of shell damage (Figure S1). As with mussel abundance and shell lengths, the amount of shell degradation also varied among sites (one-way ANOVA using the GzLM procedure, $df = 8$, $p < 0.001$). Unlike post hoc tests on abundance and shell lengths in which penultimate and ultimate marginal sites, respectively, were different from other sites, the degree of shell degradation at two centre sites associated with high densities (Elandsbaai in South Africa and Mittag in southern Namibia) differed from all other sites.

DISCUSSION

Although abundant-centre distribution is not a universal biogeographic rule across taxa (Baldanzi et al., 2013; Dallas et al., 2017; Freeman & Beehler, 2018; Sagarin & Gaines, 2002a), it was qualitatively observed in the current distributions of *Semimytilus patagonicus* in South Africa (first detected in 2009) and southern Namibia (first detected in 2014). Measurements of densities and biomass of this invasive species in South Africa descriptively fitted this pattern of distribution in 2010 (de Greef et al., 2013) and 2014 (Alexander et al., 2015) with densities centred around the centre of its range. Monitoring in early 2020 showed that the species had expanded its range northwards and south-eastwards (Ma, McQuaid, et al., 2020) with relatively low densities at marginal sites (K.C.K. Ma, personal observation), and density measurements made in 2021 (the present study) supported the presence of an abundant-centre pattern in South Africa. In southern Namibia, per cent cover measurements of the newer invasion of rocky shores by *S. patagonicus* (the present study) exhibited an overall abundant-centre pattern, assuming that the species spread from Mittag where it was first detected in the region in 2014. Thus, the distributional patterns in both South Africa and southern Namibia are quantitatively consistent with the abundant-centre distribution predicted by the ACH. The

tendency that invasive species present an abundant-centre distribution may be a common feature because they are not at equilibrium in their novel environment (Chevalier et al., 2021).

The marine mussel, *Mytilus galloprovincialis* Lamarck, 1819, has invaded the rocky shores of South Africa and qualitatively presents an abundant-centre distribution at coarse spatial scales coupled with geographic fluctuations in abundance at finer scales (Ma, Gusha, et al., 2021). Despite the existence of smaller-sized marginal populations, correspondingly lower genetic diversity towards their range edge was not found in either the invasive *M. galloprovincialis* or the native mussel, *Perna perna* (Linnaeus, 1758) (Ntuli et al., 2020). Moreover, the long-distance dispersal abilities of these species (larvae can disperse over distances greater than 100 km; McQuaid & Phillips, 2000; Gilg & Hilbish, 2003; Gilg et al., 2007) could result in higher gene flow from central to marginal populations and, in turn, breakdown of the genetic patterns predicted by the ACH. Intriguingly, marginal populations may not necessarily be located at range edges and may instead be found in regions within a species' range where (1) abundance is fluctuating greatly and at risk of local extirpation and (2) abundance is low and the species is physiologically stressed (Chardon et al., 2015; Pironon et al., 2015; Soulé et al., 1973). Overall, there has been limited support (between 31 and 39% of cases examined in review papers) for abundant-centre distributions based on abundance information, such as density (Abeli et al., 2014; Sagarin & Gaines, 2002a). From a review of plant studies, 15% of the studies exhibited higher abundance at marginal boundaries and 54% did not exhibit any difference between centre and marginal sites (Abeli et al., 2014).

By 2021, high levels of recruitment were observed (1) in South African populations at centre sites and a warm range edge site at Hermanus and (2) in the recently identified populations in southern Namibia. Further examination of South African populations revealed that densities were moderately high at the absolute range limits (i.e. not at the penultimate range edge sites), which contradicts the simple pattern of an abundant-centre distribution. Instead, the moderate upticks in abundance at range limits suggest that this is an alternative distributional pattern, such that this localized 'abundant-margin' distribution, i.e. elevated abundance at species' range limit with decreasing abundance towards the range centre, is as follows: (i) not a real phenomenon and high abundances at range edge sites are an artefact of the natural variation in abundance for an abundant-centre distribution (Figure 4a–c); (ii) a topological phenomenon of range edges for species undergoing geographic spread and this pattern will shift to new range edges as the species undergo spread (Figure 4d–f); (iii) a temporary phenomenon of range edges due to spatial heterogeneity of abundance, i.e. the current range edges happen to be marginally more optimal habitats for the species (Chardon et al., 2015; Pironon et al., 2015; Soulé et al., 1973) or a consequence of variable recruitment densities among populations (Reaugh-Flower et al., 2011; Figure 4g–i); or (iv) an occasional phenomenon of range edges due to saltatory patterns of spread resulting in the accumulation of competent larvae and high local recruitment events before they breach weak biogeographic barriers to spread or encounter a strong barrier that limits further spread (Lyons & Scheibling, 2009; Mineur et al., 2010; Ma, Gusha, et al., 2021; Figure 4j–l).

Moderately elevated *S. patagonicus* densities from the absolute range edge sites (i.e. ultimate marginal populations) could be an artefact of the natural variation in abundance for an abundant-centre distribution (Figure 4a–c), especially as the magnitudes of these values are not comparable to those from central populations. Although the observed pattern of local 'abundant-margin' distribution at both the cold and warm range edges

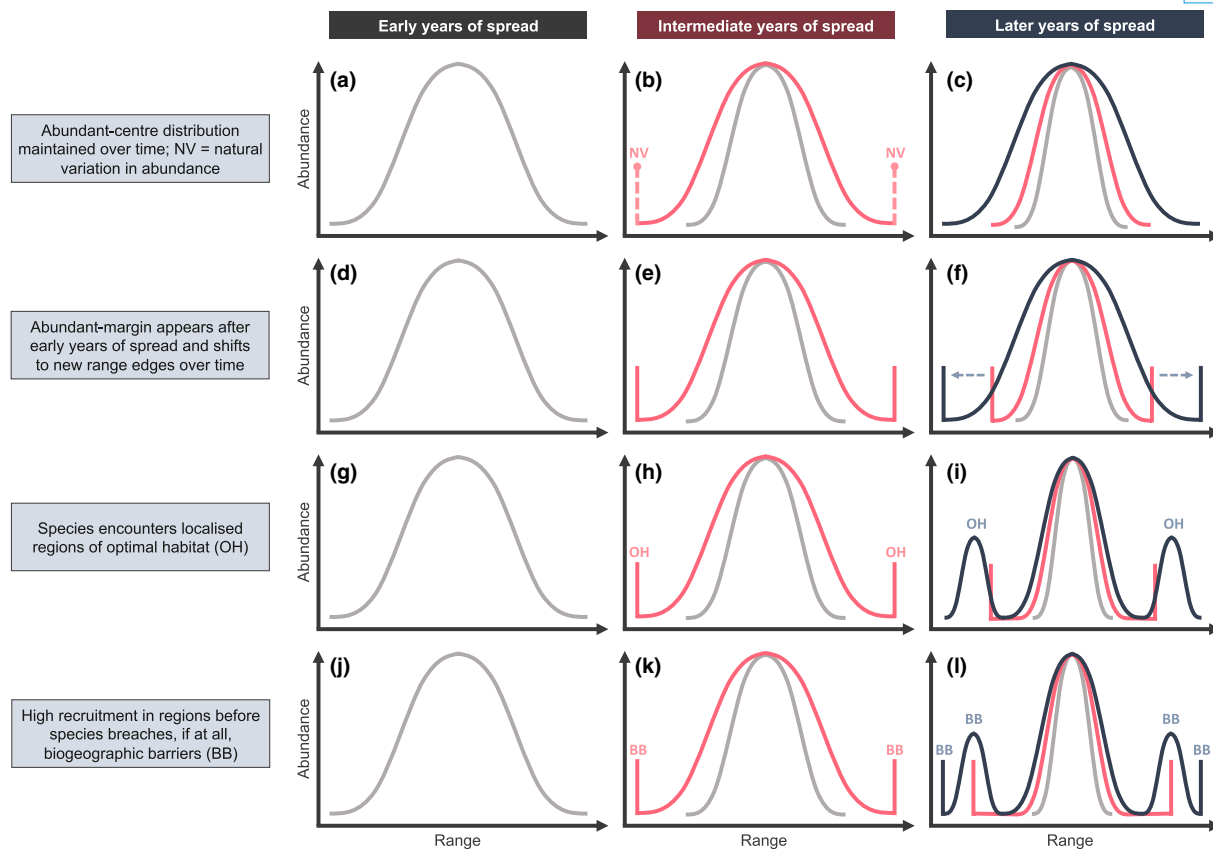


FIGURE 4 Four scenarios of the geographic spread of an invasive species (assuming a single introduction event) with a putatively abundant-centre distribution during the early years of spread and followed by the appearance of a moderate increase in abundance at the range limits (i.e. ‘abundant-margin’ pattern): (a–c) abundant-centre distribution maintained across the entire range over time and elevated abundances at range edges are just an artefact of the natural variation (NV) in abundance; (d–f) once abundant-margin patterns appear, it is a stable feature of certain species undergoing range expansions; (g–i) abundant-margin patterns are only temporary because the species is expanding into regions of optimal habitat (OH) and (j–l) abundant-margin patterns appear occasionally when the species interacts with weak/strong biogeographic barriers (BB) to spread. Note that abundant-margin patterns are depicted to appear concurrently at both range limits but that may not always be the case.

of its range may be a coincidence, the precipitous increase in densities in these ultimate marginal populations from 2020 to 2021 suggests that it was more likely a real demographic phenomenon. Whether this was an isolated occurrence or will represent a lasting ecological effect for these marginal populations remains unclear.

With sufficiently reliable presence and absence data, the sudden appearance of an ‘abundant-margin’ distribution, localized at both the cold and warm range edges of its range in 2021, may represent a distributional pattern that is a stable feature of species undergoing range expansions (Figure 4d–f). In this scenario, the abundant-margin pattern would be maintained and shift to new range edges as the species’ range expands. However, as this pattern has not been noted for the leading edges of other range-expanding taxa, this would require confirmation through long-term studies and may be hindered by correct taxonomic identification of the spreading species (e.g. spat of *S. patagonicus* can be mistaken for a spat of *M. galloprovincialis*). Instead, leading edges should eventually be characterized by smaller-sized and spatially fragmented populations as species expand towards their ecological limits and, in some cases, hybridize with other species (Bridle & Vines, 2007). The pattern of decreasing population sizes towards range limits was observed in the *M. galloprovincialis* invasion at its warm range edge on the south coast (Ma, Gusha, et al., 2021). A pattern of spatially fragmented populations was also observed at the leading

warm edge of the *S. patagonicus* invasion as its range expanded south–south-east from Bloubergstrand (also see Ma, McQuaid, et al., 2020).

In a more probable scenario, the observed ‘abundant-margin’ distributions of *S. patagonicus* could represent a temporary phenomenon as it expands into regions of optimal habitat to the north and south. Although it would be a coincidence that both the cold and warm range edges spread into the respective optimal regions concurrently, if true, the spread should then continue rapidly until the species encounters new ecological limits that slow its expansion. Although Alexander et al. (2015) predicted that *S. patagonicus* will invade the south coast of South Africa (i.e. east of Cape Agulhas), the spatial patchiness of occurrence across the south–south-eastern (and northern) portion of its range suggests the possibility of local extirpation, especially at sites characterized by small-sized populations (Soulé et al., 1973). Vectors such as hull-fouling and contaminated aquaculture products have been suggested for the introductions of *S. patagonicus* (Bigatti et al., 2014; de Greef et al., 2013; Mead et al., 2011), which can introduce new individuals to establish populations in optimal habitats outside the range of natural dispersal. Given the spatial disjunct at the warm range edge, human-mediated ‘jump’ dispersal, rather than natural range expansion (e.g. Tolley et al., 2008), could explain its isolated presence at Hermanus and its vicinity. Such processes could explain the limited support for ACH because this suggests that abundant-centre distributions in invasive species gradually break down as their ranges expand.

In another plausible scenario, the geographic spread of *S. patagonicus* may occasionally encounter weak or strong barriers to spread, such as regions of upwelling (Teske et al., 2011), regions heavily influenced by coastal currents and eddies (Assis et al., 2015; Nicastro et al., 2019; Teske et al., 2008; Zardi et al., 2011) or regions of abrupt temperature change (Bolton & Anderson, 1990; Schils & Wilson, 2006). Furthermore, models revealed the presence of a depth-specific biogeographic barrier in southern Namibia (i.e. the Lüderitz upwelling cell and Orange River cone [LUCORC] area) such that particles (e.g. larvae) are transported offshore when they are released at sub-surface depths due to hydrodynamics and environmental barriers but travel along the shore when they are released at the sea surface (Lett et al., 2007). For species undergoing range expansion, interactions with such barriers to dispersal could affect their rates of spread (Ma, Gusha, et al., 2021) and their population dynamics (e.g. through the influence of hydrodynamics on the retention and advection of planktonic larvae; Johnson, 1985; Morgan & Fisher, 2010; Morgan et al., 2011). At the cold range edge on the west coast of South Africa, larvae may have been retained in regions approaching the upwelling cells (e.g. locally at Hondeklipbaai due to the steep continental shelf; Jury & Taunton-Clark, 1986) and advected offshore once the species enters regions of persistent upwelling cells (e.g. the intensity of upwelling increases north of Hondeklipbaai; Kirkman et al., 2020). Additionally, the presence of upwelling cells on the west coast may have substantially slowed the spread of *S. patagonicus* (i.e. about 7.5 km of shore per year; Ma, McQuaid, et al., 2020). At the warm range edge of *S. patagonicus*, seasonal upwelling at Hermanus occurs from December to March, with intensity peaking in February and March (Largier et al., 1992), which may also affect the retention and advection of larvae. Furthermore, the species may be spreading towards an ecological limit on the south coast that may be dynamically changing in response to human-mediated climate change. For instance, a sudden range shift of about 73 km east of its historical range limits was detected in the macroalga, *Ecklonia maxima* (Osbeck) Papenfuss, 1940, along the south coast, in response to a gradual cooling in the region (Bolton et al., 2012).

Limited data indicate that the distribution in abundance during the early years of *S. patagonicus* spread in South Africa was consistent with the

general pattern of abundant-centre distribution (Alexander et al., 2015; de Greef et al., 2013). Also, our limited data from southern Namibia reveal a pattern of monotonic decline in per cent cover of populations from Mittag accords with the pattern of abundant-centre distribution. Moreover, the species may have reached an ecological limit given that it persisted at low abundances at the range edge at Hostel for several years (i.e. in 2017 and 2018; Pulfrich, 2018; Ma, McQuaid, et al., 2020) and, by today, has disappeared from this locality (the present study). Overall, this pattern of abundant-centre distribution is compelling, assuming a single introduction event (i.e. once in South Africa in 2009 or earlier and once in southern Namibia in 2014 or earlier) to find a source population from which the species spreads.

Although the circumstances surrounding the invasion of southern Namibia by *S. patagonicus* are largely unknown, it was probably the result of a secondary introduction via human-mediated activities or long-distance dispersal of larvae (i.e. a secondary dispersal; O'Loughlin & Green, 2017) from a previously invaded locality. We speculate that larvae were likely transported northwards from a South African source population by means of northwards littoral drift along the west coast of South Africa (de Decker, 1988; Roger & Bremner, 1991).

Unlike the newer invasion of southern Namibia, the current distribution and abundance of the ongoing invasion of South Africa by *S. patagonicus* are consistent with an abundant-centre distribution when the data from the absolute range limits (i.e. ultimate range edge populations) are excluded. When the full complement of data is viewed, the emergent pattern consists of an abundant-centre distribution coupled with moderate increases in abundance at absolute range limits (i.e. localized abundant-margin pattern), which probably represents a distributional pattern indicative of the intermediate years of spread. Plausible scenarios to explain this distributional pattern (i.e. abundant-centre with abundant-margin distribution) include either a pre-existing abundant-centre with range limits expanding into regions of optimal habitats resulting in a sudden uptick in abundance at the range edges, or a pre-existing abundant-centre with interactions (e.g. retention of competent planktonic larvae) with weak/strong biogeographic barriers to spread at the range limits. In the long-term, both these scenarios would result in a temporary abundant-margin distribution until the species spread into another relatively optimal habitat or approaches a biogeographic barrier to spread.

We observed differences in recruitment densities and extent of shell degradation between the cold and warm range edge populations. Recruitment densities were greater in warm range edge populations, whereas shell degradation was greater in cold range edge populations. It is unclear if this pattern is simply driven by thermal regimes that characterize the different marine biogeographical provinces (ecoregions) of South Africa because the warm range edge is currently located in the warm-temperate waters of the Agulhas ecoregion and the cold range edge in the cool-temperate waters of the Namaqua ecoregion (Lombard et al., 2004; Spalding et al., 2007). The higher recruitment densities at the warm range edge, coupled with the spatial patchiness of species occurrence south of the range centre, suggest that the species may be in a habitat with relatively optimal conditions. The greater degree of endolith-induced shell degradation in cold range edge populations of *S. patagonicus* is consistent with the environment-driven pattern of higher levels of endolithic infestation in cooler, nutrient-rich waters than in warmer, oligotrophic waters observed in other mussel species (Ndhlovu et al., 2019).

In conclusion, monitoring and demographic study provided insights into the range edge dynamics of an invading mussel species. For invasive species associated with a single introduction event, expectations of abundant-centre

distributions may be met during the early years of their spread, but this expectation might break down as the species expands into novel optimal habitats or towards biogeographic barriers. The recent detection and monitoring of *S. patagonicus* in southern Namibia provide a rare opportunity to examine an invasion during its early years of spread, while equivalent data at fine spatio-temporal scales are limited for other invasions. Using data from the decennial invasion of South Africa by *S. patagonicus*, we suggest that 'abundant-margin' distribution (also see Abeli et al., 2014) is a possible topological feature of the range edges of an expanding species, moderating the predictions of the ACH as a biogeographic rule (Sagarin & Gaines 2002a; Baldanzi et al., 2013; Dallas et al., 2017; Freeman & Beehler, 2018). Furthermore, both abundant-centre and an abundant-margin distributions can be resolved at the scale examined in the present study, the former distributional pattern being more dominant than the latter. This highlights the importance of sampling across the entire distributional range (e.g. Zardi et al., 2015). In order to capture patterns at relevant ecological scales and document the geographic spread of invading species over the long term and to capture temporal dynamics beyond the conceptual limits of ACH, this should include absolute range limits as in the present study (Abeli et al., 2014).

AUTHOR CONTRIBUTIONS

Kevin C. K. Ma: Conceptualization (lead); data curation (equal); formal analysis (lead); investigation (lead); methodology (equal); project administration (equal); visualization (lead); writing – original draft (lead); writing – review and editing (equal). **Andrea Pulfrich:** Data curation (equal); funding acquisition (equal); methodology (equal); project administration (equal); writing – review and editing (equal). **P William Froneman:** Funding acquisition (equal); supervision (supporting); writing – review and editing (equal). **Christopher D McQuaid:** Funding acquisition (equal); supervision (lead); writing – review and editing (equal).

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DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.